

PROPOSAL FOR VIRTUAL EXHIBIT

1.Title

"Clockwork: The Rhythm of the Processor"

2. Group's Member Roster

Student Name	Student ID	Role
Nathan Laborada	12414808	Front-end development and Exhibit Design and research
Azzam Alonto	12301272	Research, Content Writer, QA, Dev
Bryan Choa	12411027	Front-end development and Exhibit Design
Daniel Angelo Gonzalez	12306940	Back-end development
Keion Clei Chua	12343110	Research, Content Writer, Back-end development

3. Group's Topic Theme

"Tick, Tock, Execute: Following the Clock Through the CPU Pipeline"

The CPU clock is the fundamental metronome that synchronizes every single operation inside the processor. Our exhibit walks visitors through the whole journey, from the physical quartz crystal inside the computer all the way to how the clock orchestrates the fetch-decode-execute cycle. We think it's a nice way to connect the hardware side with the conceptual side (pipelining and performance).

4. Group's Tech Stack Plan

Layer	Technology	Purpose
Framework	Astro 6 with React.js	The spec requires Astro, and React gives us the interactivity we need for the simulations
Styling	Tailwind CSS	Fast to work with, and we can match the museum template's look without fighting with custom CSS
Core Animation	HTML5 Canvas and Framer Motion	Canvas is great for the wave rendering. Framer Motion

		handles the UI animations and button feedback
Data Visualization	D3.js	We only need it for the wave chart and performance graphs. Keeping it lightweight

5. Proposed Interactive Elements

This exhibit features two interactive elements that work together to tell the complete story of the CPU clock.

Interactive Element #1: "Clock Speed Simulator"

Feature	Description
Clock Speed Slider	Drag it from 1 MHz up to 5 GHz. The wave and everything else responds instantly .
Square Wave Visualizer	Renders the live clock signal on a Canvas. At low speeds you can see each tick clearly. At high speeds it turns into a blur.
Heat Gauge	A thermometer-style indicator showing estimated CPU temperature. As frequency increases, temperature rises exponentially (demonstrating the power wall: $\text{Heat} \propto \text{Voltage}^2 \times \text{Frequency}$).
Voltage Display	Shows how voltage must increase to sustain higher frequencies, contributing to the heat problem.
Cycle Counter	A running counter showing how many clock cycles have elapsed since the visitor started the simulation.

Real-time Feedback:

When you move the slider, the wave speeds up or slows down immediately. Below the wave, we show a simple assembly instruction (like ADD RAX, RBX) with a progress bar. At 1 GHz, you can actually watch it execute tick by tick. At 5 GHz, it finishes in a flash. This makes the abstract concept of clock speed feel tangible.

Interactive Element #2: "Pipeline Visualizer"

Feature	Description
5-Stage Pipeline Diagram	A Gantt-chart-style visualization showing IF, ID, EX, MEM, and WB stages.
Sample Program Loader	Pre-loaded programs with 4–5 instructions (e.g., MOV, ADD, SHL, ROR from your quiz).

Run / Step Controls	Users can click "Run" to watch the program execute automatically, or "Step" to advance one clock cycle at a time.
Pipeline Stalls (Hazards)	A "Hazard" toggle button. When enabled, a data hazard is introduced (where one instruction depends on the result of the previous one). The pipeline visually "stalls" and "bubbles" appear, showing why pipelining isn't perfect.

Real-time Feedback:

Colored blocks slide forward one stage per clock tick. A performance counter shows total cycles taken with and without the hazard enabled. Hovering over a stage reveals a tooltip explaining what happens there (for example, "EX: ALU computes the arithmetic operation").

6. Mobile-Responsive Layout

Breakpoint	Width	Layout Structure
Desktop	≥1024px	Three-panel layout: Left (20% - Navigation & Glossary), Center (50% - Main Visuals: Clock Wave + Pipeline), Right (30% - Controls: Slider, Run Button, Performance Stats).
Tablet	768px - 1023px	Two-panel layout: Top (Controls + Stats), Bottom (Main Visuals with tabs to switch between Clock Wave and Pipeline views).
Mobile	<768px	Single-column vertical stack: Clock Speed Simulator on top, Pipeline Visualizer in the middle, Controls and Performance Stats at the bottom. All touch targets are at least 44px for easy interaction.

7. Tentative Style Guide Snapshot

7.1 Color Palette ("Oscilloscope / Lab Equipment" Theme)

Element	Color Code	Usage
Background	#0A0E17	Main dark canvas (like a lab oscilloscope screen).
Surface Cards	#141B2B	Panel backgrounds.

Primary Accent (Clock Wave)	#00FFC8	Neon Cyan/Mint - for the square wave and active data buses.
Secondary Accent (Heat/Power)	#FF6B6B	Neon Red/Coral - for the heat gauge and power wall warnings.
Tertiary Accent (Pipeline)	#4ECDC4	Teal - for pipeline stage blocks.
Text Primary	#E8EDF2	Off-white for body text.
Text Secondary	#8892B0	Muted slate for metadata and labels.



7.2 Typography

Element	Font	Weight
Headers / Titles	JetBrains Mono	Bold (700)
Clock Frequency / Values	JetBrains Mono	Normal (400)
Body / Descriptions	Inter	Regular (400)

7.3 Layout Wireframe (Desktop)

The following ASCII wireframe shows the three-panel desktop layout:

LEFT PANEL (20%): Navigation menu with sections for Intro, Clock Wave, Pipeline, Heat, and Performance.

CENTER PANEL (50%): Main visual stage with animated square wave, instruction execution progress bar, and 5-stage pipeline diagram with colored stage blocks.

RIGHT PANEL (30%): Interactive controls including clock speed slider (1 MHz - 5 GHz), voltage display, heat gauge thermometer, cycle counter, Run/Step buttons, hazard toggle, and performance metrics display.

8. References

Intel 64 and IA-32 Architectures Software Developer's Manual

Computer Organization and Design (Patterson & Hennessy)

React.js Documentation

Tailwind CSS Documentation

Github Repo: https://github.com/chewsdycat/CSARCH2_GRP2.git